

Euclid's Elements — Book V

Proposition 13

Formal Reconstruction — Technical Documentation

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CGJTEAMLAB

If a first ratio is equal to a second, and the second is greater than a third, then the first is also greater than the third.

1 Introduction

Proposition V.13 is a transport theorem for strict ratio comparison. It combines equality of ratios, expressed by Eudoxus' Definition 5, with the strict greater-ratio relation represented by a witness in Definition 7.

The proof is short, but its role in the Book V dependency graph is important. It is the bridge that allows a strict comparison manufactured by V.8 to be moved through an existing equality of ratios before V.10 converts the transported ratio comparison back into an inequality of magnitudes.

The production file is

```
CGJteamLab/Proposition5_13.lean
```

and it imports only

```
CGJteamLab.HilbertInterfaceV
```

from the project-level Book V infrastructure.

2 Formal Statement of the Comparison Pattern

Assume

$$a : b = c : d$$

and

$$c : d > e : f.$$

Then V.13 asserts

$$a : b > e : f.$$

The first hypothesis is an equality of ratios in the sense of Definition 5. The second is a strict ratio comparison in the sense of Definition 7.

The proposition therefore transports an already existing strict witness through an equality of ratios.

3 The Definition 7 Witness

The hypothesis

$$c : d > e : f$$

means that there exist positive-multiple indices m, n such that

$$M_n(d) < M_m(c)$$

and

$$\neg(M_n(f) < M_m(e)).$$

The crucial observation is that the second clause already has exactly the form needed for the target

$$a : b > e : f.$$

Only the first clause must be transported.

4 Transport through Definition 5

From

$$a : b = c : d,$$

Definition 5 gives, for the same indices m, n ,

$$M_n(b) < M_m(a) \iff M_n(d) < M_m(c).$$

The witness hypothesis contains the right-hand side:

$$M_n(d) < M_m(c).$$

Therefore the equivalence immediately yields

$$M_n(b) < M_m(a).$$

Together with the unchanged negative clause

$$\neg(M_n(f) < M_m(e)),$$

the same pair m, n witnesses

$$a : b > e : f.$$

No new multiple indices are constructed.

5 Formal Statement

The production theorem is:

```

theorem euclid_proposition_5_13
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (a b c d e f : HilbertPositiveSegmentClass Geo)
  (hProp :
    HilbertEudoxusProportion Geo a b c d)
  (hGreater :
    HilbertEudoxusGreaterRatio Geo c d e f) :
  HilbertEudoxusGreaterRatio Geo a b e f := by

rcases hGreater with
  \langle m, n, hCD, hNotEF\rangle

refine
  \langle m, n, ?_, hNotEF\rangle

exact
  (hilbertEudoxusProportion_gt_iff
    Geo
    a b c d
    hProp
    m n).2
  hCD

```

The entire mathematical proof consists of reusing the same witness m, n and transporting one strict comparison.

6 The Extraction Lemma

The reusable interface lemma is

```
hilbertEudoxusProportion_gt_iff
```

which extracts the reverse-strict-comparison clause of Definition 5.

For fixed m, n and a proportion

$$a : b = c : d,$$

it states

$$M_n(b) < M_m(a) \iff M_n(d) < M_m(c).$$

In V.13 the implication is used from right to left, since the strict witness is initially attached to $c : d$.

7 Lean Proof Architecture

The proof DAG is

$$\begin{array}{c}
 \text{hGreater} : c : d > e : f \\
 \Downarrow \\
 \exists m, n, M_n(d) < M_m(c) \wedge \neg(M_n(f) < M_m(e)) \\
 \Downarrow \\
 \text{hProp} : a : b = c : d \\
 \Downarrow \\
 M_n(b) < M_m(a) \iff M_n(d) < M_m(c) \\
 \Downarrow \\
 M_n(b) < M_m(a) \\
 \Downarrow \\
 M_n(b) < M_m(a) \wedge \neg(M_n(f) < M_m(e)) \\
 \Downarrow \\
 a : b > e : f.
 \end{array}$$

No case split and no order trichotomy is needed.

8 Axiomatic Strength

The theorem requires only

```
[HilbertIncidence Geo]
[HilbertCongruence Geo]
```

through the positive segment-class infrastructure.

In particular, V.13 does not use

`HilbertArchimedeanPlane`

and does not invoke the least-exceeding-multiple theorem.

The Archimedean work needed to create a greater-ratio witness belongs to V.8. V.13 merely transports a witness that has already been supplied.

9 Relationship with V.8

V.8 produces strict ratio comparisons from strict magnitude comparisons. For example,

$$a < c$$

can be turned into a comparison of ratios sharing a common second term.

V.13 then allows that strict ratio comparison to pass through an equality

$$a : b = c : d.$$

This is the middle step required in the strict part of V.14.

Schematically,

$$\begin{array}{c} a < c \\ \Downarrow \text{V.8} \\ c : d > a : d \\ a : b = c : d \\ \Downarrow \text{V.13} \\ a : b > a : d. \end{array}$$

V.10 then reads the final strict ratio comparison back as

$$b < d.$$

10 Relationship with V.10 and V.14

V.13 by itself does not produce an inequality between magnitudes. Its conclusion is still a strict comparison of ratios.

The transition back to magnitude order is V.10:

$$a : b > a : d \implies b < d.$$

Thus the strict-comparison mechanism of V.14 factors naturally as

$$V.8 \longrightarrow V.13 \longrightarrow V.10.$$

This factorization is explicit in `Proposition5_14.lean`.

11 Hidden Formal Data

The proof has one important hidden datum: the witness indices m, n .

Euclid's prose treats the strict ratio comparison abstractly. In the Lean definition, however, the hypothesis

```
HilbertEudoxusGreaterRatio Geo c d e f
```

contains explicit existential data.

The proof preserves that exact pair of indices. There is no second existence argument and no need to synchronize different witnesses.

This is why V.13 is so short formally.

12 Classical DAG versus Formal DAG

At the ratio level, the proposition is simply:

$$\begin{array}{c} a : b = c : d \\ c : d > e : f \\ \Downarrow \\ a : b > e : f. \end{array}$$

At the witness level:

$$\begin{array}{c} M_n(d) < M_m(c) \\ \neg(M_n(f) < M_m(e)) \\ \Downarrow \\ M_n(b) < M_m(a) \iff M_n(d) < M_m(c) \\ \Downarrow \\ M_n(b) < M_m(a) \\ \neg(M_n(f) < M_m(e)) \\ \Downarrow \\ a : b > e : f. \end{array}$$

The formal DAG therefore exposes the single comparison transported through Definition 5.

13 Structural Lesson

V.13 illustrates the value of separating equality of ratios from strict comparison of ratios.

Definition 5 supplies a family of comparison equivalences. Definition 7 supplies one witness pair m, n . V.13 combines them without any new magnitude construction.

This division of responsibilities is important for the later propositions: V.8 is responsible for existence, V.13 for transport, and V.10 for order reflection.

14 Conclusion

Proposition V.13 is the transport rule for strict ratio comparison through Eudoxean equality of ratios.

Its proof reuses the same Definition 7 witness and transports only the positive comparison clause through Definition 5. The negative clause is unchanged.

Although formally small, V.13 is the essential middle node of the proof graph

$$\text{V.8} \longrightarrow \text{V.13} \longrightarrow \text{V.10} \longrightarrow \text{V.14},$$

and therefore an important structural component of the Book V layer.